

Entry Sections and Relation-Valued Descent

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2026

This paper (Paper XXVII of the RIME program) develops fixed-scope entry-section and relation-valued descent interfaces for single-defect circular automata.

Abstract

Problem. In the fixed $n=6$ and $n=7$ single-defect circular settings considered here, local rank descent is not determined by the mass state alone. Already at rank four, reachable states in these fixed scopes need not admit the desired local escape, and selecting one shortest word per mass endpoint can erase source-addressed packet information needed by the proof.

Approach. We replace a universal checkpoint predicate by an intrinsic entry section and replace a deterministic continuation witness by a future-free, relation-valued completion interface. The resulting proof objects retain typed ancestry only to the level at which the required source-addressed relation is proved to descend.

Results. In the fixed $n=6$ single-defect cycle family, an intrinsic section selects one legitimate rank-four checkpoint for each of the 1,704 synchronizing rooted rank-five defects. A computer-assisted classification, validated against the bound source closure, gives 1,700 Type-I-admitting instances and four genuinely Type-II-only instances, the latter forming two explicit repayment mechanisms. For the inherited extremal $n=7$ carrier, we prove a Low-Transport theorem by finite symbolic exhaustion. Its canonical finite elimination consists of one empty shallow cell, 17 typed post-tag states, 40 kinematic returns reduced to eight admitted returns, and five rank-three exit relations containing 17 admissible heavy-pair exits.

Boundary. These are fixed-scope structural theorems. We do not prove the Černý conjecture, a general First-Fusion Selector theorem, or an all- n uniform bound on relation-menu size.

Keywords. synchronizing automata; Černý conjecture; mass quotient; entry section; typed ancestry; relation-valued descent; finite proof; provenance

Notation Table

symbol	meaning
$Q = \mathbb{Z}/n\mathbb{Z}$	finite state set in the declared circular scope
p	fixed cyclic permutation letter
d	collision-rooted rank- $(n - 1)$ defect letter
$A = (Q, \{p, d\})$	rooted single-defect circular action
$\mu_t = t_* \mathbf{1}$	image-fiber mass of a transformation t
$M(\mu), \tau(\mu)$	collision mass and maturity potential
$S(\mu \rightarrow \mu')$	endpoint-normalized corridor surplus
$P_{\leq 3}$	exact nonrecursive terminal base $P_1 \cup P_2 \cup P_3$
$C_r = (A, d, \mu_r, \chi_r)$	typed source-rank- r context
$\Sigma_r^{(n)}$	future-free selector from source rank r in ambient size n
$\mathcal{E}_6^{\text{entry}}, \pi_6, \tilde{\sigma}_6$	admissible-entry relation, its action projection, and the chosen right inverse
$\kappa(B)$	intrinsic ordering key for an admissible macro block B
$\text{Im}^{\text{idx}}(\Sigma)$	selector image retaining its source index
$\mathcal{R}_4^{(7)}(X)$	complete source-addressed completion relation at checkpoint X
$s_4^{(7)}(X)$	deterministic serialization of that relation
$\mathcal{E}(Y)$	admitted endpoint-shortest heavy-pair exit words from Y
$\Gamma(Y, t)$	passive image of $\mathcal{E}(Y)$ under t
$\mathbf{N}, \mathbf{A}$	endpoint-normalization and accounting admission predicates

1 Introduction

Let p be a fixed cycle and d a rank- $(n - 1)$ defect letter with one binary kernel. The mass quotient records how transformation fibers merge, while corridor compression records the rank-preserving transport required before a strict fusion. This framework converts a reset route into a sequence of local fusion corridors with exact length accounting.

The first obstruction is semantic. Already in the fixed single-defect cycle scopes studied here, a rank-four mass state may be reachable in the semigroup without being a legitimate checkpoint for the recursive proof. Even activated ancestry does not make every rank-four checkpoint locally escapable. The fixed $n=6$ solution is therefore not another state predicate: it is a section that chooses one legitimate activated checkpoint from each synchronizing rooted action.

The second obstruction is representational. At $n=7$, two tied shortest words can reach the same mass endpoint while transporting different labelled packet identities. A quotient that retains only the endpoint is therefore too coarse for a moved-slot relation. At the opposite extreme, retaining the complete action-resolved digest audited here merely re-encodes the finite context. The useful interface lies between these extremes: a finite, future-free relation menu with an existentially certified descending member.

The two fixed-rank results support one organizing sentence:

$n = 6$ chooses a legitimate checkpoint; $n = 7$ determines the relation structure it must retain.

1.1 From Mechanism Classification to Proof-State Architecture

A mechanism-first organization would classify balanced and exceptional repayment patterns, prove descent into $P_{\leq 3}$, and then seek a higher-rank return. The finite obstructions below show that this classification is downstream of a prior semantic question: which proof state makes local descent well-defined?

The mass state is too coarse to certify a legitimate recursive checkpoint. Typed activation restores ancestry but still does not choose a descending representative, which forces the intrinsic entry section of Section 3. After that choice has been made, a second quotient failure occurs: tied endpoint-shortest words can carry different source-addressed packet transports. Retaining the particular complete action-resolved digest audited here, however, merely re-encodes the finite context. The resulting progression of proof objects is therefore

$$\boxed{\begin{array}{l} \text{legitimate checkpoint choice} \longrightarrow \text{endpoint-quotient failure} \\ \longrightarrow \text{source-addressed relation} \longrightarrow \text{sufficient finite descent interface} \end{array}} \quad (1.1)$$

This is a conceptual progression, not a logical dependency between the fixed- $n=6$ and fixed- $n=7$ theorems. Local descent mechanisms remain necessary realizations inside the interface, but their taxonomy and recursive composition remain outside the present theorem spine.

Terminology boundary. *Relation-valued descent* here means a finite, source-addressed completion relation attached to one typed proof context. Paper XXIV instead studies patch-indexed relation families, natural-join reconstruction, and the α -acyclicity obstruction [5]. The two papers share a descent philosophy, not a mathematical object or theorem.

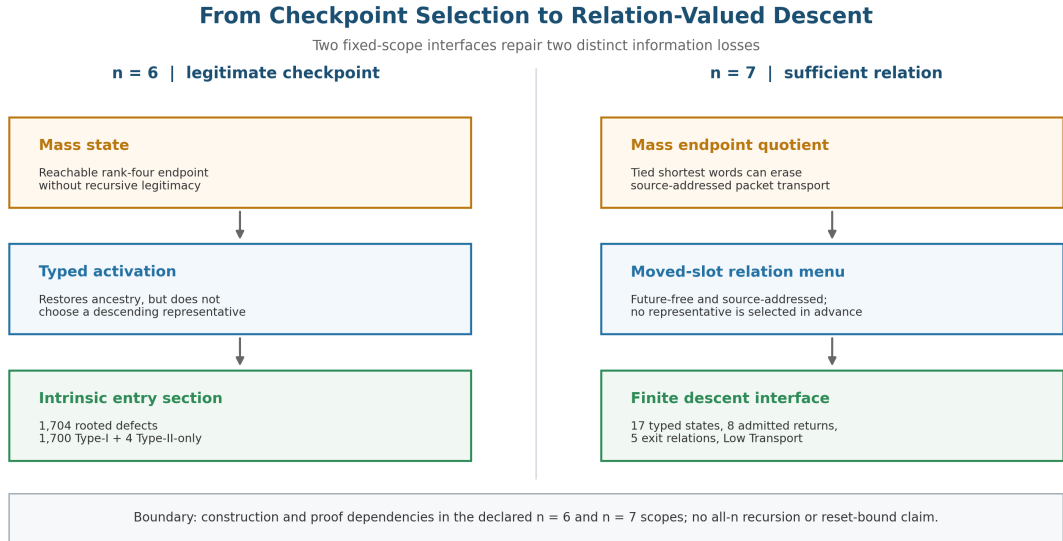


Figure 1. The fixed- $n=6$ section chooses a legitimate checkpoint; the fixed- $n=7$ relation interface preserves the source-addressed transport needed for descent. Each arrow denotes a within-scope construction dependency; the diagram asserts neither an implication from the $n=6$ theorem to the $n=7$ theorem nor an all- n recursion.

1.2 Contributions

This paper makes four fixed-scope contributions.

1. **Legitimate checkpoint selection.** It defines an intrinsic $n=6$ entry section and proves that state gates, typed activation, and section choice have different logical roles.

2. **Quotient failure and source addressing.** It shows that tied endpoint-shortest representatives may carry different packet-addressed transports, so the required relation does not descend through the mass endpoint quotient.
3. **A sufficient proof interface.** It isolates the fixed-scope contract used here: bounded, future-free, relation-valued, and descent-certified, with provenance closure imposed separately as a publication requirement.
4. **Fixed-scope realizations.** The $n=6$ indexed section graph has 1,700 Type-I-admitting and four Type-II-only instances, while the inherited extremal $n=7$ carrier satisfies Low Transport through a complete finite symbolic elimination.

1.3 Nonclaims

No statement below asserts any of the following:

- that every rank-four mass state admits local descent;
- that activation alone guarantees descent;
- that every member of a relation menu succeeds;
- that a successful member is unique;
- that the observed fixed-scope menu bounds are uniform in n or rank;
- that the fixed $n=6, 7$ results prove Forced FFS, General FFS, or the Černý conjecture.

The design objective is to reduce the retained proof state without re-encoding the full context; no mathematical minimality claim is made.

2 Mass, Corridors, and Two-Corridor Accounting

Words act from left to right. Thus, for a mass state μ and a letter a ,

$$\mu_{ta} = a_*\mu_t.$$

For a transformation $t : Q \rightarrow Q$, define its image-fiber mass by

$$\mu_t(y) = |t^{-1}(y)|.$$

Its rank and collision mass are

$$r(\mu) = |\text{supp}(\mu)|, \quad M(\mu) = \sum_y \binom{\mu(y)}{2}.$$

For a nonuniform mass state of rank r , define the maturity potential

$$\tau(\mu) = \|\mu\|_2^2 + (r-1)(n-r) - 2n + 1,$$

with the separate convention $\tau(\mathbf{1}) = 0$ at the uniform initial mass.

A corridor from rank r to rank $r - q$ has endpoint-normalized length ℓ . Writing $\Delta M = M(\mu') - M(\mu)$, its surplus is

$$S(\mu \rightarrow \mu') = \tau(\mu') - \tau(\mu) - \ell = 2\Delta M + q(2r - q - n - 1) - \ell. \quad (2.1)$$

Along a reset route the surpluses telescope:

$$\sum S = (n - 1)^2 - \text{total word length.} \quad (2.2)$$

The two-corridor rule permits exactly two macro types.

- **Type-I:** one corridor with $S \geq 0$.
- **Type-II:** two corridors with $S_1 < 0$ and $S_1 + S_2 \geq 0$.

The exact low-rank sets P_1, P_2, P_3 form the terminal base used here; this base is not defined recursively. We write

$$P_{\leq 3} = P_1 \cup P_2 \cup P_3.$$

The distinction between a mass endpoint and a source-addressed corridor is essential. Endpoint normalization is imposed on complete words for a fixed endpoint; it does not authorize quotienting tied shortest representatives when their packet-addressed transports differ.

3 Fixed n=6: An Intrinsic Entry Section

Let

$$Q = \mathbb{Z}/6\mathbb{Z}, \quad p(i) = i + 1.$$

Let d be a collision-rooted rank-five defect with binary kernel K_d and collision image 0. Let m be the unique coordinate omitted from $\text{im}(d)$. The kernel-image mass $\mu_d = d_* \mathbf{1}$ has mass two at 0, mass zero at m , and mass one elsewhere.

Define

$$a(d) = \min\{a \in \{0, 1, 2\} : p^a(m) \notin K_d\}. \quad (3.1)$$

The minimum exists because $m, p(m), p^2(m)$ are distinct while K_d has two points.

Lemma 3.1 (short-entry normal form).

Every endpoint-shortest strict block from μ_d has the form

$$p^{a(d)}d, \quad a(d) \in \{0, 1, 2\}.$$

Proof. *If the current rank-five support is $Q \setminus \{x\}$ with $x \in K_d$, then a non-strict application of d satisfies*

$$d(Q \setminus \{x\}) = \text{im}(d) = Q \setminus \{m\}. \quad (3.2)$$

Thus every non-strict d resets the support to the same canonical support. In a strict word

$$p^{a_0}d p^{a_1}d \cdots p^{a_k}d,$$

all non-strict initial segments can be deleted while preserving strictness of the final segment. A shortest strict word therefore contains one copy of d , and (3.1) gives $a \leq 2$. QED.

Every such short entry is activated and Type-I: at rank five its length is at most three and its first fusion has nonnegative surplus.

Definition 3.2 (predecessor idempotent).

Set

$$\text{PredIdem}(d) \iff d^2 = d \quad \text{and} \quad K_d = \{0, p^{-1}(0)\}.$$

Define the intrinsic section

$$\sigma_6(d) = \begin{cases} p^5 d, & \text{PredIdem}(d), \\ p^{a(d)} d, & \text{otherwise.} \end{cases} \quad (3.3)$$

The exceptional clause recognizes predecessor-collapse geometry; it does not recognize an action string or query future escape.

Proposition 3.3 (rooted scope classification).

The 1,800 collision-rooted binary rank-five defects split exactly into

$$24 \mathcal{A}\text{-preserving} \sqcup 72 \mathcal{E}\text{-preserving} \sqcup 1 \text{ PredIdem} \sqcup 1703 \text{ generic defects}, \quad (3.4)$$

where

$$\mathcal{A} = \{\{0, 3\}, \{1, 4\}, \{2, 5\}\}, \quad \mathcal{E} = \{\{0, 2, 4\}, \{1, 3, 5\}\}.$$

The first two classes preserve a nontrivial quotient and are not synchronizing. In this exact fixed scope, the remaining 1,704 defects are synchronizing. The converse direction in this sentence is a finite classification, not an all-n block-system theorem.

Let $\mathcal{S}_6$ denote these 1,704 synchronizing rooted defects. Counts for the section are taken in its indexed graph

$$\text{Im}^{\text{idx}}(\sigma_6) = \{(d, \text{end}(\sigma_6(d))) : d \in \mathcal{S}_6\}.$$

Thus the action index is retained even when two actions have the same mass endpoint. The ordinary endpoint projection of this graph is not what is counted below.

To make the word section literal, define the admissible-entry relation

$$\mathcal{E}_6^{\text{entry}} = \{(d, w) : d \in \mathcal{S}_6, w \text{ is an admissible activated } 5 \rightarrow 4 \text{ entry for } d\}, \quad \pi_6(d, w) = d. \quad (3.4a)$$

The selected lift

$$\tilde{\sigma}_6(d) = (d, \sigma_6(d)), \quad \pi_6 \circ \tilde{\sigma}_6 = \text{id}_{\mathcal{S}_6} \quad (3.4b)$$

*is therefore a choice section of π_6 . It is unrelated to the **Option**-valued signature sections of Paper XXIV [5].*

Lemma 3.4 (predecessor zero-surplus comb at n=6).

For the predecessor idempotent, repeated use of $p^5 d$ produces

$$(2, 1, 1, 1, 1, 0) \xrightarrow{p^5 d} (3, 1, 1, 1, 0, 0) \xrightarrow{p^5 d} (4, 1, 1, 0, 0, 0) \xrightarrow{p^5 d} (5, 1, 0, 0, 0, 0) \xrightarrow{p^5 d} (6, 0, 0, 0, 0, 0),$$

with zero surplus in every block. The endpoint-shortest $p^2 d$ entry is an activated hostile control, but

$$(d, \text{end}(p^2 d)) \notin \text{Im}^{\text{idx}}(\sigma_6).$$

Theorem 3.5 (fixed- $n=6$ entry-section descent).

For every synchronizing collision-rooted binary rank-five defect d , the word $\sigma_6(d)$ is an activated rank-five-to-rank-four entry, and its endpoint admits a local Type-I or Type-II macro descent into $P_{\leq 3}$.

Partition the indexed graph into its Type-I-admitting and genuinely Type-II-only parts. More precisely,

$$\text{Im}^{\text{idx}}(\sigma_6) = \text{Im}_I^{\text{idx}}(\sigma_6) \sqcup \text{Im}_{I \setminus I}^{\text{idx}}(\sigma_6), \quad (|\text{Im}_I^{\text{idx}}(\sigma_6)|, |\text{Im}_{I \setminus I}^{\text{idx}}(\sigma_6)|) = (1700, 4). \quad (3.5)$$

The four Type-II-only indexed instances form two mechanisms, each occurring twice:

source partition	first fusion	repayment fusion
(2, 2, 1, 1)	$1 + 2 \rightarrow 3$	$3 + 2 \rightarrow 5$
(3, 1, 1, 1)	$1 + 1 \rightarrow 2$	$3 + 2 \rightarrow 5$.

(3.6)

Both terminate at partition $(5, 1)$ in the exact set P_2 . Their total surpluses are respectively

$$14 - (\ell_1 + \ell_2) \quad \text{and} \quad 12 - (\ell_1 + \ell_2).$$

The theorem quantifier is existential over legal receipts. It does not require the canonically serialized receipt to have the displayed common form.

The executable enumeration establishes the finite classification and the **1700+4** indexed-graph decomposition. The bound certificate records its theorem-facing output, while the release validator checks source binding and closure. The symbolic equalities supplied in the proof remain mathematical premises rather than conclusions inferred from the certificate files.

Remark 3.6 (what the section repairs).

The fixed-scope hierarchy is strict:

$$\text{state gate} < \text{typed activation} < \text{typed section choice}. \quad (3.7)$$

State-local rank-four escape has genuine counterexamples. Activation removes counterfactual placements but still leaves activated endpoints without local descent. The intrinsic section succeeds on all 1,704 synchronizing rooted actions. Thus activation is necessary for semantic legitimacy; in this complete fixed scope it is not sufficient for descent, whereas the section choice is.

4 From Deterministic Witnesses to Relation-Valued Proof Objects

Definition 4.1 (source-rank convention and inherited section chain).

For ambient size n , a typed source-rank- r context is written

$$C_r = (A, d, \mu_r, \chi_r),$$

where $A = (Q, \{p, d\})$ is the rooted action, μ_r has rank r , and χ_r records the labelled packet placement, activation ancestry, and source roles needed by the local admissible relation. The subscript of $\Sigma_r^{(n)}$ is always the **source rank**, while the superscript is the ambient cardinality. Thus a selected block

$$B = \Sigma_r^{(n)}(C_r)$$

has rank- $r-1$ endpoint, and the endpoint context $\text{End}(C_r, B)$ inherits the endpoint mass, labelled packets, and the fresh/parent packet data of the selected fusion. In particular,

$$\Sigma_5^{(6)} = \sigma_6, \quad \Sigma_6^{(7)} : 6 \rightarrow 5, \quad \Sigma_5^{(7)} : 5 \rightarrow 4.$$

For an admissible source-addressed macro block $\mathbf{B}$, let

$$\kappa(B) = (\ell(B), \text{type}(B), w_1(B), w_2(B), \rho(B), \mu_{\text{end } B})$$

in lexicographic order, where Type I precedes Type II and $\text{rho}(\mathbf{B})$ is the ordered boundary-parent coordinate tuple of the first fusion. This is an intrinsic serialization of the local relation, not a downstream success key.

For a rooted synchronizing defect at $n = 7$, start with

$$C_6(d) = (A_d, d, \mu_d, \chi_6(d)), \quad \mu_d = d_* \mathbf{1},$$

where $\chi_6(d)$ is the labelled kernel-image packet placement. The two future-free selectors are defined as follows.

1. **Rank-six selector.** $\Sigma_6^{(7)}$ forms all activated, endpoint-normalized Type-I blocks from μ_d to rank five and selects the κ -least block. For the intrinsic predecessor idempotent only, it selects the declared oriented return $p^6 d$.
2. **Rank-five selector.** First form

$$C_5(d) = \text{End}(C_6(d), \Sigma_6^{(7)}(C_6(d))).$$

Then $\Sigma_5^{(7)}$ forms the complete endpoint-normalized admissible macro relation, restricts it to blocks ending at rank four, and selects its κ -least member. Every block in this restricted relation is Type I: a Type-II block contains two strict drops and bypasses rank four.

Thus the inherited chain is

$$C_6(d) \xrightarrow{\Sigma_6^{(7)}} C_5(d) \xrightarrow{\Sigma_5^{(7)}} C_4(d).$$

Both choices read only the rooted action, current mass, typed ancestry, and the current local admissible relation. Neither selector reads

$$L_4^{(7)}, \quad P_{\leq 3} \text{ membership, target winning, } W^{2C}, \\ \text{reset coaccessibility, or a Bellman policy.}$$

The predecessor correction occurs only in $\Sigma_6^{(7)}$.

Let $\mathcal{C}_5^{(7)}$ be the complete finite set of inherited contexts $C_5(d)$ generated by this first selector in the declared $n = 7$ scope. The second selector defines the indexed graph

$$\text{Im}^{\text{idx}}(\Sigma_5^{(7)}) = \{(C, \text{end}(\Sigma_5^{(7)}(C))) : C \in \mathcal{C}_5^{(7)}\}.$$

The selector is defined on all 15,120 context-indexed instances; projected rank-four endpoints are not asserted to be distinct. This graph is the image of the checkpoint selector only; all of its selected blocks have one strict drop and are Type I in the source-rank sense above.

For an indexed checkpoint

$$X = (C, C_4) \in \text{Im}^{\text{idx}}(\Sigma_5^{(7)}),$$

let $\mathcal{R}_4^{(7)}(X)$ be the complete source-addressed relation of legal Type-I and Type-II completion receipts from C_4 into $P_{\leq 3}$. The source index C is retained, so equal projected rank-four mass endpoints are not silently identified. The declared deterministic serialization is

$$s_4^{(7)}(X) = \text{Serialize}(\mathcal{R}_4^{(7)}(X)).$$

It is a shortest-witness view of the relation, not a selector in the inherited section chain or the theorem quantifier. Across the 15,120 indexed checkpoints, this serialization chooses 14,936 Type-I receipts and 184 Type-II receipts.

The quantifier-correct existential split is instead induced by the complete completion relation. Define

$$\begin{aligned}\mathcal{G}_I^{(7)} &= \{X : \mathcal{R}_4^{(7)}(X) \text{ contains a Type-I receipt}\}, \\ \mathcal{G}_{II \setminus I}^{(7)} &= \{X : \mathcal{R}_4^{(7)}(X) \text{ contains no Type-I receipt but contains a Type-II receipt}\}.\end{aligned}$$

Then

$$\text{Im}^{\text{idx}}(\Sigma_5^{(7)}) = \mathcal{G}_I^{(7)} \sqcup \mathcal{G}_{II \setminus I}^{(7)}, \quad (|\mathcal{G}_I^{(7)}|, |\mathcal{G}_{II \setminus I}^{(7)}|) = (15093, 27). \quad (4.1)$$

This discrepancy is not cosmetic. A selected witness is a serialization choice; the theorem object is the complete legal receipt relation.

We use *completion relation* for the complete legal receipt relation and *relation menu* for the finite, theorem-facing family of future-free role channels exposed to a descent argument. A role channel may contain more than one concrete receipt; the existential descent statement ranges over the receipts represented by the menu.

Common-witness audits support an intermediate representation. Coarse signatures do not determine one completion template, while the particular complete local-relation digest used in the audit is injective on all 15,120 contexts and therefore supplies no compression on this scope. Intermediate labelled classes nevertheless admit finite observed covers. Those broader counts are pre-release discovery records, not theorem claims, definitions, or members of the public release identity.

The extremal carrier of Sections 5–8 is not an independently sampled rank-four census. It is the 35-instance cell of the indexed checkpoint graph in which both selected section blocks are p^2d , the inherited rank-five partition is $(2, 2, 1, 1, 1)$, $\Sigma_5^{(7)}$ fuses $1 + 1$ with length three and surplus zero, and the resulting rank-four partition is $(2, 2, 2, 1)$ with the declared F_4/K_d carrier incidence. Sections 5–8 study $\mathcal{R}_4^{(7)}$ on this cell; Section 5 first reconstructs the carrier intrinsically.

Definition 4.2 (proof-interface contract).

The relation $\mathcal{R}_r(C)$ below is context-local: it is attached to one typed proof context C and ranges over source-addressed completion receipts. It is neither a patch family nor a natural-join object, and no α -acyclicity theorem is invoked. Those objects belong to the distinct typed-context descent problem of Paper XXIV [5].

A fixed-scope completion interface assigns each typed rank- r context C a finite relation menu. Write $\mathcal{R}_r(C)$ for its realized completion relation: the set of all concrete receipts represented by the menu. The interface is required to be:

property	mathematical contract
Bounded	each declared context has a finite menu; no all-n uniform constant is asserted
Future-free	construction reads no downstream success, winning, Bellman, or reset-coaccessibility label
Relation-valued	tied representatives and packet identities are retained whenever the observable does not descend to a quotient
Descent-certified	at least one admissible menu member descends

Provenance closure is a separate publication contract: implementation, bound artifacts, and source closure must agree. Any performed producer replay must also agree, and replay status is recorded explicitly.

The descent condition is deliberately existential:

$$\exists B \in \mathcal{R}_r(C) : B \text{ is admissible and descends.} \quad (4.2)$$

It neither requires every menu member to succeed nor chooses a unique winning channel.

Remark 4.3 (lossy endpoint quotient).

Let $W_{\min}$ be the complete fixed-endpoint shortest-word relation and let

$$q : W_{\min} \rightarrow \mathcal{E}$$

forget the word while retaining its mass endpoint. For a source-addressed observable $\mathbf{R}$, the implication

$$q(w_1) = q(w_2) \implies R(w_1) = R(w_2) \quad (4.3)$$

is false in the n=7 carrier. Tied shortest words can reach the same mass endpoint while fusing different packet identities. The correct theorem object is therefore the fixed-endpoint shortest-word relation, not one BFS representative per endpoint.

This correction simplifies the canonical relation profile to

$$72 = 61_{\text{direct}} + 5_{\text{one-return}} + 1_{\text{OC}} + 5_{\text{bad}}. \quad (4.4)$$

The numbers in (4.4) are verification data, not the Low-Transport theorem.

5 The Extremal n=7 Carrier

Let

$$Q = \mathbb{Z}/7\mathbb{Z}, \quad p(i) = i + 1, \quad K_d = \{0, 6\},$$

and write

$$d = (0, \pi(1), \dots, \pi(5), 0), \quad \pi \in S_5.$$

Consider the inherited extremal rank-four carrier of partition $(2, 2, 2, 1)$ created by two p^2d returns. Its local action is parameterized by

$$(h, \alpha, \beta) \in \{3, 4, 5\} \times S_3 \times S_2. \quad (5.1)$$

Set

$$\lambda(1) = 1, \quad \lambda(2) = 2, \quad \lambda(3) = h, \quad \alpha(j) = \pi(\lambda(j)).$$

Lemma 5.1 (carrier reconstruction).

The triple (h, α, β) uniquely determines π . Explicitly,

$$\pi(1) = \alpha(1), \quad \pi(2) = \alpha(2), \quad \pi(h) = \alpha(3). \quad (5.2)$$

If $\{u, v\} = \{3, 4, 5\} \setminus \{h\}$ with $u < v$, then

$$\begin{aligned} \beta = I : \quad & (\pi(u), \pi(v)) = (4, 5), \\ \beta = S : \quad & (\pi(u), \pi(v)) = (5, 4). \end{aligned} \quad (5.3)$$

The ambient 36 carrier actions are therefore exactly

$$\mathcal{C}_{\text{amb}}^{(7)} = \{3, 4, 5\} \times S_3 \times S_2.$$

Define the predecessor boundary and selected extremal carrier by

$$\mathcal{C}_{\text{pred}}^{(7)} = \{(3, e, I)\}, \quad \mathcal{C}_{\text{sel}}^{(7)} = \mathcal{C}_{\text{amb}}^{(7)} \setminus \mathcal{C}_{\text{pred}}^{(7)}. \quad (5.4)$$

Thus the 36 oriented parameter cells are the disjoint union of the 35-context selected carrier and one predecessor section exception. Sections 6–8 analyze $\mathcal{C}_{\text{sel}}^{(7)}$; the excluded predecessor cell is retained only as an adjacent boundary case.

Lemma 5.2 (packet recursion).

Let F_4 be the packet created by the selected rank-five-to-rank-four return, let D_1, D_2 be the other mass-two packets, and let s be the singleton. Their rooted coordinates are

$$\boxed{F_4 = 0, \quad D_1 = \alpha(2), \quad D_2 = \pi(\alpha(2) + 2), \quad s = \pi(\alpha(3) + 2).} \quad (K)$$

All coordinates are taken modulo seven.

Definition 5.3 (moved-slot relation).

For $j \in \text{Mov}(\alpha)$, let $W_j(C)$ be the complete relation of role-compatible, endpoint-normalized heavy-comb associations in which the incoming mass-two packet traverses the tagged edge

$$\lambda(j) \xrightarrow{d} \alpha(j)$$

during its own fusion corridor. Define $\text{Land_1}(W_j)$ to be the subrelation whose final singleton offset is one. The definition is future-free and retains all tied endpoint-shortest representatives.

6 Canonical Five-Cell Elimination

The Low-Transport proof uses positive constructor soundness everywhere and complete relation exhaustion only at the five parameter cells where a universal negative statement is required. This section packages that finite elimination as five theorem-facing equalities.

6.1 Typed shallow states

A post-tag state is

$$\widehat{X} = (X_{\text{place}}, c, \text{incoming role}, k, \delta, \text{labelled corridor source}), \quad (6.1)$$

where c is the corridor index, k is the marked-prefix length, and δ is the accounting state needed by later admission. Prefixes may be quotiented only after they induce the same typed state and the same normalization and accounting semantics.

The five menu-bad parameter cells are

$$\mathfrak{B} = \{(132, 2, 3, I), (213, 2, 3, I), (213, 2, 3, S), (213, 2, 4, I), (321, 3, 3, I)\}. \quad (6.2)$$

Here α is written in one-line notation on the three low slots.

Lemma 6.1 (empty shallow cell).

For direct and one-return germs, equivalently post-tag depth $r \in \{0, 1\}$,

$$\boxed{\mathcal{X}_3^{\text{sh}}(321; h = 3, \beta = I) = \emptyset.} \quad (6.3)$$

The two corridor roles fail for different reasons. A first-corridor mark makes every candidate violate the two-corridor accounting bound $\ell_1 + \ell_2 \leq 20$; a second-corridor mark violates fixed-endpoint minimality. No landing test is used.

*The superscript **sh** is essential. The full tagged relation is nonempty and contains a depth-two endpoint-shortest representative. Thus (6.3) is an empty canonical shallow-menu relation, not an empty full relation.*

Lemma 6.2 (back-solving exhaustion).

For the four nonempty cells in (6.2), solving the typed reverse constraints gives

$$\boxed{\bigsqcup_b \text{Sol}_{BT}(b) = \{\widehat{X}_1, \dots, \widehat{X}_{17}\}.} \quad (6.4)$$

The reverse calculation starts from the terminal fusion roles, uses unique reverse rotations, and branches under d^{-1} only at the collision image. Every rejected branch has one of four declared reasons:

$$\begin{aligned} E &= \text{early fusion}, & R &= \text{role violation}, \\ N &= \text{fixed-endpoint nonminimality}, & A &= \text{debt or length violation}. \end{aligned}$$

Exactly one displayed state, $\widehat{X}_5$, has two tied provenances; they may be identified because they induce identical typed placement and gate semantics. This is a valid quotient, in contrast to (4.3). The marked-prefix count is retained in Appendix B as an exhaustion checksum rather than a theorem object.

Lemma 6.3 (return admission).

For a typed state $\widehat{X}$, let $Q^{\text{kin}}(\widehat{X})$ be the rotations satisfying the rank-preserving and role-preserving kinematic equations. Let $N(\widehat{X}, q)$ and $A(\widehat{X}, q)$ denote fixed-endpoint normalization and accounting admission. Then, row by row,

$$\boxed{Q(\widehat{X}_i) = \{q \in Q^{\text{kin}}(\widehat{X}_i) : N(\widehat{X}_i, q) = A(\widehat{X}_i, q) = 1\}.} \quad (6.5)$$

On this five-cell surface, normalization and accounting are the complete admission interface. The displayed rows contain both accounting-only and normalization-only rejections, so neither gate is implied by the other. The aggregate rejection counts are recorded in Appendix B as transcription checks. Equality (6.5), not those counts, is the theorem.

Lemma 6.4 (heavy-pair exit exhaustion).

The first-corridor branches in (6.4)-(6.5) produce five typed rank-three states Y_k . Define

$$\mathcal{E}(Y_k) = \{w : w \text{ is an admissible endpoint-normalized strict heavy-pair exit from } Y_k\}. \quad (6.6)$$

Then $\mathcal{E}(Y_k)$ equals the complete displayed word set in the finite exit table. The shortest-parent DAG first exhausts fixed-endpoint shortest words, after which the heavy-pair role and debt filters

are applied. Appendix B records the complete admitted/over-debt/wrong-pair partition as a completeness checksum; the word-set equality (6.6) is the theorem-facing statement.

For a passive singleton at coordinate $\mathbf{t}$, define the image relation

$$\boxed{\Gamma(Y_k, t) = \{w(t) : w \in \mathcal{E}(Y_k)\}}. \quad (6.7)$$

The five exact images are

$$\begin{aligned} \Gamma(Y_1, 4) &= \{2, 4, 5\}, \\ \Gamma(Y_2, 5) &= \{2, 4, 5\}, \\ \Gamma(Y_3, 5) &= \{3, 4, 5\}, \\ \Gamma(Y_4, 5) &= \{2, 3, 5\}, \\ \Gamma(Y_5, 5) &= \{3, 5\}. \end{aligned} \quad (6.8)$$

No singleton image is used to choose an exit word; (6.8) is the passive image of the already exhausted relation (6.6).

Corollary 6.5 (five negative spectra).

The complete direct and one-return landing spectra at the cells in $\mathfrak{B}$ are

cell	direct spectrum	one-return spectrum
132/2, (3, I)	{2, 4, 5}	{2, 4, 5}
213/2, (3, I)	{3, 4, 5}	{4}
213/2, (3, S)	{4}	empty
213/2, (4, I)	{2, 3, 4, 5}	{3, 5}
321/3, (3, I)	empty	empty

Offset one is absent in every row, and the independent Orbit-Closure germ is inapplicable at all five parameter cells. The corollary is a set-image consequence of Lemmas 6.1-6.4; it bears no independent search or completeness burden.

Appendix B displays the complete 17-row back-solving table, the rowwise 40-to-8 admission table, and the five complete heavy-pair word sets. Their equality proofs, not the counts alone, are the mathematical premises.

7 Low Transport

Call a moved direction j menu-good if a sound direct, one-return, or Orbit-Closure constructor supplies a member of $\mathbf{W}_j(\mathbf{C})$ landing at offset one. Constructor soundness gives only the implication needed below:

$$\text{MenuGood}_j \implies \text{Land}_1(W_j) \neq \emptyset. \quad (7.1)$$

No converse or normal-form theorem for all deep successful germs is used.

Lemma 7.1 (transposition landing).

The five-cell elimination yields

$$\begin{aligned} B_2(132) &\iff h = 3 \wedge \beta = I, & B_3(132) &\iff \perp, \\ B_1(213) &\iff \perp, & B_2(213) &\iff h = 3 \vee (h = 4 \wedge \beta = I), \\ B_1(321) &\iff \perp, & B_3(321) &\iff h = 3 \wedge \beta = I. \end{aligned} \quad (7.2)$$

Thus every transposition has at least one good moved direction:

$$\alpha \text{ a transposition} \implies \exists j \in \text{Mov}(\alpha) : \text{Land}_1(W_j) \neq \emptyset. \quad (7.3)$$

The existential quantifier is sharp: five transposition contexts have only one good direction.

Lemma 7.2 (three-cycle landing).

If $\alpha = 231$, every moved direction has a direct offset-one constructor. The same holds for $\alpha = 312$ except at $(h, \beta, j) = (5, S, 2)$, where the explicit Orbit-Closure germ supplies the relation. Hence

$$\alpha \text{ a three-cycle} \implies \forall j \in \{1, 2, 3\}, \quad \text{Land}_1(W_j) \neq \emptyset. \quad (7.4)$$

The Orbit-Closure word closes the tagged packet's four-cycle before the final strict fusion. A separate Root-Shift Exchange identity remains valid but is not a dependency of the canonical proof after tied-shortest correction.

Theorem 7.3 (Low-Transport Lemma).

Let C be a non-predecessor extremal carrier context. If α is nonidentity, then

$$\boxed{\exists j \in \text{Mov}(\alpha), \quad \exists B \in W_j(C) : \text{Land}(B) = 1.} \quad (7.5)$$

Equivalently, $\alpha \neq e$ implies $1 \in R_4(C)$.

Proof. Every nonidentity element of S_3 is a transposition or a three-cycle. Apply Lemma 7.1 or Lemma 7.2. QED.

8 The Identity Boundary and Credit Reallocation

When $\alpha = e$, the moved-slot menu is empty. The ambient 36-cell table has six such cells: five belong to $\mathcal{C}_{\text{sel}}^{(7)}$, while the predecessor cell belongs to $\mathcal{C}_{\text{pred}}^{(7)}$.

h	β	π	selected outcome or boundary status
3	I	e	adjacent boundary: predecessor section exception
3	S	$(4 \ 5)$	offset-one heavy comb
4	I	$(3 \ 4)$	offset-two heavy-comb bridge
4	S	$(3 \ 5 \ 4)$	offset-one heavy comb
5	I	$(3 \ 4 \ 5)$	offset-one heavy comb
5	S	$(3 \ 5)$	rank-two $(5, 2)$ fallback

The predecessor cell is outside the selected carrier and uses the independent p^6d section. The $(3 \ 4)$ cell uses an explicit p^3dp^2d bridge. The $(3 \ 5)$ cell uses the rank-two $(5, 2)$ fallback.

For the source partition $(2, 2, 2, 1)$,

$$\tau(2, 2, 2, 1) = 9, \quad \tau(7) = 36.$$

Thus the total maturity credit is 27. The heavy-comb and fallback channels allocate it as

$$\boxed{20 + B_2(6, 1) = 20 + 7 = 27, \quad 12 + B_2(5, 2) = 12 + 15 = 27.} \quad (8.1)$$

The $(5, 2)$ fallback does not create credit. It spends eight fewer units in the rank-four prefix and leaves eight additional units for the rank-two tail.

Theorem 8.1 (extremal carrier completion at $n=7$).

Every context in the 35-context selected carrier $\mathcal{C}_{\text{sel}}^{(7)}$ admits a local completion into the exact low-rank base. Its 30 cells with $\alpha \neq e$ are covered by Theorem 7.3. Its five non-predecessor cells with $\alpha = e$ are covered by three direct heavy-comb channels, one bridge, and one (5, 2) fallback. Equation (8.1) closes the two possible terminal credit allocations.

The predecessor cell $\mathcal{C}_{\text{pred}}^{(7)}$ is outside the theorem domain. It is recorded in the ambient boundary table and handled independently by the p^6d section.

This theorem is a human-auditable finite proof with source-bound computational verification. It is not an all- n return theorem.

9 Sufficient Proof Interfaces

The fixed-rank results supply a sufficient proof-interface contract and suggest the following proof-design principle:

Proof-interface principle. A quotient is admissible only after the theorem-relevant observable is shown to descend through it.

Equivalently, compress proof-relevant information only through quotients on which the required relation is proved to descend. The design objective is to reduce the retained proof state without re-encoding the full context; no mathematical minimality claim is made.

At $n=6$, the mass state is too coarse to choose a legitimate checkpoint; typed activation and an intrinsic section are needed. At $n=7$, one shortest word per mass endpoint is too coarse for packet-addressed transport, while the particular full action-resolved digest audited here is injective and therefore too fine to compress this scope. The theorem uses typed states and set-valued relations at the intermediate level.

Schematically, on the audited fixed scopes,

$$\boxed{\text{too-coarse endpoint quotient} \quad | \quad \text{sufficient interface} \quad | \quad \text{injective full digest.}} \quad (9.1)$$

This display juxtaposes two proved counterexamples to overcompression with one observed noncompressing full digest. It defines no order on representations and makes no minimality claim for the middle interface.

The contrast between two quotients is instructive.

- The endpoint quotient is forbidden because the moved-slot relation does not descend through it.
- The two-prefix quotient at $\widehat{X}_5$ is allowed because both provenances induce the same typed state and all later gate semantics agree.

Thus *lossless* is always relative to the theorem observable. The goal is not to retain every available field, nor to force a deterministic outcome, but to retain exactly enough structure to prove an existential descent channel.

9.1 Audited proof-state refinements

The interface was not selected for notational convenience. Each refinement below is forced by a declared hostile audit.

hostile audit	failed proof object	information retained	canonical replacement
reachable	mass state	typed ancestry	activated checkpoint
rank-four failures			
activated	activated checkpoint	legitimate source	intrinsic entry section
rank-four failures		choice	
tied-shortest	mass-endpoint quotient	source-addressed	fixed-endpoint
failure		packet transport	shortest-word relation
full-digest	the particular complete	theorem-relevant	finite future-free relation
over-resolution	action-resolved digest	channels only	menu
	audited here		

Thus every enlargement of the proof state is paid for by a counterexample to the preceding quotient. The last row has a deliberately narrower status: the full digest is injective on the declared finite context population and hence offers no useful compression there. It does not prove that every lossless local encoding must be injective or uninformative.

9.2 Mechanisms are realizations, not interfaces

Balanced repayments, two-corridor fallbacks, bridges, heavy combs, and orbit closures remain necessary ingredients of the finite proofs. They describe how a selected local channel succeeds, and may vary with the local realization. Checkpoint legitimacy and descent of the source-addressed relation instead control whether such mechanisms can be composed without changing the proof's meaning. The present paper therefore treats local mechanisms as realizations inside a sufficient proof interface, not as the interface itself.

The all-rank candidate is a finite-valued section correspondence

$$\Sigma_r(C_r) = \{\mathcal{C}_1(C_r), \dots, \mathcal{C}_m(C_r)\},$$

constructed without future success information and satisfying

$$\exists j : \text{end}(\mathcal{C}_j(C_r)) \in \text{Sec}_{r-1}. \quad (9.2)$$

Neither a uniform bound on m nor an all-rank definition of $\text{Sec}_{\{r-1\}}$ is proved here.

10 Related Work and Novelty Boundary

This section separates inherited class results, adjacent proof methods, and the fixed-scope interface claimed here. The comparison is by hypotheses, mathematical object, and proof mechanism; shared generator format alone does not identify the present theorem surface.

10.1 Circular and one-cluster automata

The automata studied here contain a letter acting as a full cycle. They therefore lie inside the classical circular-automaton setting, for which the Černý bound is already known by Dubuc [6]. The numerical reset bound is not a novelty claim of this paper. Prime-cycle one-cluster automata were treated by Steinberg [12], and Zhu's recent annular-spectral preprint proves the Černý bound for one-cluster automata in its stated positive-level setting [16]. Those works establish reset-length results by extension or linear-algebraic methods; they do not supply the fixed-rank entry-section and source-addressed relation interfaces proved here.

10.2 Defect-one actions, groups, and reachable subsets

Automata generated by permutation letters together with a rank- $n-1$ merging letter are a recognized structural and computational class. Catalano and Jungers study randomized generation and slowly synchronizing families in this format [4]. Casas Torres studies complete reachability for almost group automata with exactly one defect-one letter and a set of permutation letters [3]. Rystsov proves the Černý and rank conjectures for the narrower Černý-type setting generated by a simple idempotent and a regular permutation group [10]. Synchronizing-group and transformation-monoid methods provide a broader algebraic setting for permutation actions combined with singular maps [1, 11]. The present work overlaps those sources at the alphabet and semigroup level, but its fixed cyclic action, rooted packet ancestry, endpoint-normalized corridors, and two-corridor accounting are additional hypotheses and proof objects.

Complete reachability asks whether every nonempty subset is the image of the whole state set [2, 7]. The Ferens-Szykuła journal version gives a quadratic decision algorithm and a quadratic reaching bound, while Zhu gives binary counterexamples to Don’s proposed reaching bound and a near-bound for standardized automata [7, 15]. That theory is adjacent because it organizes defect words and subset-image transport. We neither assume nor prove complete reachability. Our sections choose one proof-relevant checkpoint, and the relation menus retain only the source-addressed local channels needed for the declared descent.

10.3 Extension methods and relation-valued witnesses

Subset extension and preimage growth are central tools in synchronization. Kisielewicz and Szykuła exhibit subsets whose shortest extending words are quadratic and explain why a naive uniformly short extension principle cannot generally improve the cubic method [8]. This paper makes no universal short-extension claim. Its relation-valued interface instead preserves all tied shortest representatives when the moved-slot observable fails to descend through the mass-endpoint quotient.

Rodaro and Venturi study a different quotient question: lifting the Černý property from quotient automata via congruence and transition-monoid ideal structure [9]. Their automaton-quotient problem is distinct from the present proof-data question of whether a source-addressed observable descends through a proposed information quotient.

Recent surveys record the surrounding class results and open questions about compression, subset synchronization, and linear algebra [13, 14]. Within the cited comparison set, we did not locate an equivalent formulation of the combined interface

intrinsic entry section + future-free source-addressed completion relation.

This is a bounded attribution statement, not an exhaustive novelty certificate.

10.4 Overlap, disjoint scope, and claimed contribution

comparison surface	overlap	disjoint scope	claim retained here
circular automata [6]	a full-cycle letter	prior reset-bound theorem is global; our results are fixed-rank interface theorems	no numerical Černý-bound novelty
one-cluster and linear extension [12, 16]	cyclic transport and subset growth	different hypotheses, carrier, and proof mechanism	entry and relation interfaces only
permutation plus one rank- $n-1$ letter [3, 4, 10]	raw generator format	randomized, complete-reachability, or reset/rank-conjecture objectives	rooted indexed section-graph classification
completely reachable automata [2, 3, 7, 15]	defect words and subset images	complete reachability is neither assumed nor concluded	selected proof checkpoint and local menu
synchronizing groups [1, 11]	permutation action with a singular map	group-wide synchronization properties	fixed cyclic, source-addressed transport
extension and quotient methods [8, 9]	short words, preimage transport, and quotient structure	no universal extension bound or automaton-quotient lifting theorem is asserted	theorem-relative preservation of tied witnesses
finite typed-context descent [5]	descent language and relation-valued data	patch families, natural joins, and α -acyclicity	one-context completion relation with an existential witness

The contribution is therefore an interface architecture, not a new numerical reset-threshold theorem:

state predicate $\longrightarrow$ entry section $\longrightarrow$ relation-valued descent interface.

11 Evidence Status and Provenance

For the finite-symbolic equalities in Appendix B, the paper and artifacts have different jobs:

the paper proves the Appendix B equalities;
 bound artifacts check transcription and provenance closure;
 replay status is recorded separately.

(11.1)

Theorem 3.5 has a different status: executable enumeration establishes its complete fixed- $n=6$ classification, and the bound certificate records the theorem-facing output. The release validator checks source binding and closure. The manuscript supplies the definitions, local identities, and claim boundary for that computer-assisted theorem.

We use three epistemic labels throughout:

label	meaning in this paper
symbolic	a general formula or local identity is derived in the text without finite-scope machine exhaustion
finite symbolic exhaustion	a declared finite parameter set is exhausted by explicit equations and complete paper tables
computer-assisted classification	a complete fixed scope is classified by executable enumeration whose recorded output is checked against a bound source closure

Accordingly, Theorem 3.5 is a computer-assisted fixed-scope classification; carrier reconstruction and Theorem 7.3 use finite symbolic exhaustion; the credit identity and the explicit closed-germ identities are symbolic. The 15,120-context statistics are computational evidence and are not theorem premises.

The theorem-facing equalities are:

$$\begin{aligned}
\mathcal{X}_3^{\text{sh}}(321; 3, I) &= \emptyset, \\
\bigsqcup_b \text{Sol}_{BT}(b) &= \{\widehat{X}_1, \dots, \widehat{X}_{17}\}, \\
Q(\widehat{X}_i) &= \{q \in Q^{\text{kin}}(\widehat{X}_i) : N = A = 1\}, \\
\mathcal{E}(Y_k) &= \{\text{all displayed admissible exits}\}, \\
\Gamma(Y_k, t) &= \text{eval}_t \mathcal{E}(Y_k).
\end{aligned} \tag{11.2}$$

Counts such as 129 complete representatives, 184 canonically serialized Type-II receipts, or relation-cover statistics are discovery diagnostics. They do not replace the equalities in (11.2) and are not required by the public release closure.

12 Open Problems

The present work leaves the following questions open.

1. Determine whether relation-menu size admits a useful all-n bound.
2. Find an intermediate representation that is lossless for the required observable without becoming injective on the full context space.
3. Characterize the class of theorem observables for which a source-addressed local relation admits a noninjective sufficient quotient.

Mechanism taxonomy, parameterized repayment families, credit accounting, and future-free return between sections remain outside the present theorem spine. In particular, no n=8, Forced FFS, B2-generalization, or all-rank mechanism result is part of Paper XXVII.

Any enlargement of the present interface requires an additional typed state, an admitted return hidden by a third gate, an omitted strict exit, another failed quotient, or a performed replay that changes a theorem-facing relation.

13 Conclusion

The two fixed-rank analyses answer successive proof-interface questions:

$n = 6$ chooses a legitimate checkpoint;
 $n = 7$ determines the relation structure it must retain.

At $n=6$, the correct object is an intrinsic entry section rather than a universal rank-four predicate. At $n=7$, the correct local object is a source-addressed, relation-valued completion interface rather than one chosen shortest representative or the particular injective full digest audited here. The resulting finite proofs separate theorem-facing equalities from bound computational evidence and provenance records.

Mechanism classification remains outside the present theorem spine.

These conclusions do not supply an all-rank section, a uniform bound on menu size, Forced FFS, General FFS, or a new proof of the Černý conjecture. Their role is narrower: they identify and verify a proof architecture on the two finite scopes studied here, where checkpoint choice and inherited relation structure can be observed together.

Appendix A Claim Status

claim	status
mass/corridor/two-corridor accounting	exact framework input
fixed- $n=6$ rooted scope and section classification	computer-assisted classification
fixed- $n=6$ 1700+4 indexed section-graph theorem	computer-assisted classification
extremal $n=7$ carrier reconstruction	symbolic formulas plus finite parameterization
five-cell canonical elimination	finite symbolic exhaustion, closed at declared scope
Theorem 7.3 (Low-Transport)	finite symbolic consequence of the closed elimination
identity boundary and 27-credit closure	symbolic finite completion
all- n bounded relation menu	open
Forced FFS / General FFS	open
Černý conjecture	not claimed

Appendix B Complete Five-Cell Elimination

This appendix records the complete theorem-facing finite equalities used in Section 6. It is included for human auditability. The bound artifacts check their transcription and provenance closure; any producer replay is recorded separately and is not assumed here. The artifacts are not premises for the completeness claims.

B.1 Empty Shallow Cell

The fifth negative parameter cell is separate. Here the post-tag relation is explicitly the **shallow** relation

$$\mathcal{X}_j^{\text{sh}}(C) = \{\widehat{X} : \widehat{X} \text{ arises from a role-compatible marked } C_4 \text{ association with } r \in \{0, 1\}\}.$$

No landing test occurs in this definition. The claim to prove is

$$\boxed{\mathcal{X}_3^{\text{sh}}(321; h = 3, \beta = I) = \emptyset.} \quad (\text{Empty-X})$$

Carrier reconstruction gives

$$d = (0, 3, 2, 1, 4, 5, 0), \quad (F_4, s, D_1, D_2) = (0, 1, 2, 4),$$

and the marked moved edge is $3 \rightarrow 1$. Put $H = F_4 + D_2$ after the first heavy-comb fusion. For corridor lengths ℓ_1, ℓ_2 , the two surpluses are

$$S_1 = 7 - \ell_1, \quad S_2 = 13 - \ell_2.$$

Thus a Type-II heavy comb requires

$$\ell_1 \geq 8, \quad \ell_1 + \ell_2 \leq 20. \quad (\text{EX-A})$$

The finite back-solving used below is entirely local. Reverse rotation is unique, and the inverse fibres of the defect are

$$d^{-1}(0) = \{0, 6\}, \quad d^{-1}(1) = \{3\}, \quad d^{-1}(3) = \{1\}, \quad d^{-1}(i) = \{i\} (i = 2, 4, 5). \quad (\text{EX-B})$$

Starting from the prescribed terminal fusion roles, iterate (EX-B) through rank-preserving predecessors. Mark a predecessor exactly when the incoming mass-two packet occupies coordinate 3, and retain only marks with at most one later nonterminal d. Endpoint nonminimal branches and branches violating (EX-A) are discarded. This gives the following two exhaustive calculations.

B.1.1 A first-corridor mark cannot be repaid

Solving the first-corridor equations for both choices of incoming mass-two packet leaves exactly the five marked words below. Digits use $0 = p, 1 = d$; the bracketed 1 is the marked occurrence. Every surviving row fuses F4 with D2.

id	marked first word	r	rank-three endpoint
F1	000101[1]01000001	1	0:H,3:D1,5:s
F2	1000000[1]0100001	1	0:H,3:D1,5:s
F3	10000010[1]0100001	1	0:H,2:s,3:D1
F4	1000000[1]0000001	0	0:H,2:D1,3:s
F5	000101[1]000001	0	0:H,1:s,5:D1

Their accounting data are:

id	ℓ_1	minimum possible ℓ_2	total	exclusion
F1	15	6	21	A
F2	15	6	21	A
F3	16	6	22	A
F4	15	12	27	A
F5	13	8	21	A

The minimum possible ℓ_2 is the shortest strict second exit fusing H with D_1 , over all exact rank-two endpoints. Consequently every row violates $\ell_1 + \ell_2 \leq 20$; no first-corridor mark defines a member of $\mathcal{X}_3^{\text{sh}}$.

B.1.2 A second-corridor mark is nonminimal

The same back-solving without a mark gives exactly five first corridors that can satisfy the repayment bound:

id	complete first word	rank-three source Z	ℓ_1	maximum ℓ_2
Z1	000010001	0:H,1:D1,3:s	9	11
Z2	0000101001	0:H,2:s,5:D1	10	10
Z3	00001011001	0:H,1:D1,2:s	11	9
Z4	00010000001	0:H,1:s,4:D1	11	9
Z5	00010100001	0:H,2:s,4:D1	11	9

For a final singleton coordinate t , let

$$L_{\text{sh}}(Z, t)$$

be the minimum length, within the displayed repayment bound, of a strict $H + D_1$ exit containing a shallow marked $3 \rightarrow 1$ edge. Let $L_{\text{min}}(Z, t)$ be the unrestricted fixed-endpoint minimum. Solving the labelled pair-exit equations gives all finite shallow values:

source	$t : L_{\text{sh}}(Z, t) > L_{\text{min}}(Z, t)$
Z1	1: 9>8, 2: 9>7, 3: 9>7, 5: 10>7
Z2	5: 10>8
Z3	2: 9>8, 3: 9>7, 5: 9>7
Z4	none
Z5	none

Every omitted coordinate has no shallow marked solution within the repayment bound. Every displayed solution is strictly longer than another word to the same mass endpoint, so fixed-endpoint normalization rejects it. This proves (Empty-X) before any return or rank-three continuation calculation.

The superscript *sh* is essential. The full tagged relation is not empty: from Z_1 , the endpoint-shortest word 00[1]01100001 contains the marked edge and has two later rank-preserving d letters ($r = 2$). It is outside the canonical direct/one-return menu and is retained as a hostile control against the stronger, false untyped claim.

B.2 Back-Solving Exhaustion

An accounting triple is (S_{in}, k, b) : surplus entering the marked corridor, marked-prefix length, and running balance after the tagged occurrence.

The four nonempty carrier inputs, derived from Lemma 5.1 and (K), are:

cell	rooted defect	rank-four source	marked edge
132/2, (3, I)	0132450	0:F4,3:D1,4:s,5:D2	2 -> 3
213/2, (3, I)	0213450	0:F4,1:D1,3:D2,5:s	2 -> 1
213/2, (3, S)	0213540	0:F4,1:D1,3:D2,4:s	2 -> 1
213/2, (4, I)	0214350	0:F4,1:D1,4:D2,5:s	2 -> 1

For a corridor source Z , incoming role R , and moved slot j , a back-solving solution is a marked prefix $u=u' d$ such that:

1. every $\mathbf{d}$ in $\mathbf{u}$ is rank-preserving;
2. $\mathbf{R}$ occupies $\text{lambda}(\mathbf{j})$ immediately before the final, marked $\mathbf{d}$ in $\mathbf{u}$;
3. the resulting post-tag state admits a direct or one-return suffix of the prescribed fusion corridor;
4. the full corridor is shortest for its exact mass endpoint and satisfies the two-corridor accounting gate.

For corridor one, $\mathbf{Z}$ is the displayed rank-four source. For corridor two, $\mathbf{Z}$ ranges over the rank-three endpoints of admissible first corridors. The inverse step is explicit:

$$p^{-1}(i) = i - 1 \pmod{7}, \quad d^{-1}(0) = \{0, 6\}, \quad d^{-1}(i) = \{\pi^{-1}(i)\} \quad (i \neq 0). \quad (\text{BT-inverse})$$

Back-solving the terminal roles with **(BT-inverse)** and discarding a branch at its first $\mathbf{E}$, $\mathbf{R}$, $\mathbf{N}$, or $\mathbf{A}$ violation is a finite inverse recurrence. More explicitly, for each cell $\mathbf{b}$, corridor $\mathbf{c}$, and incoming role $\mathbf{R}$, start from the two prescribed terminal fusion parents. Reverse rotations uniquely. Whenever a reversed $\mathbf{d}$ -edge has image zero, branch over the two preimages $\{0, 6\}$; every other nonzero coordinate has a unique predecessor. Stop a branch when it reaches the marked equation

$$R@_{\lambda}(\mathbf{j}) \xrightarrow{d} R@_{\alpha}(\mathbf{j}), \quad (\text{BT-mark})$$

or at its first $\mathbf{E}/\mathbf{R}/\mathbf{N}/\mathbf{A}$ violation. Fixed-endpoint distance and the remaining debt budget bound the marked-prefix depth. Subtracting the shortest admissible post-tag suffix from those bounds gives the channel caps in the next table.

Inverse-Tree Termination and Completeness Lemma. For every declared cell and channel, the recurrence above terminates and enumerates every admissible marked prefix within its stated depth cap. Reverse rotations are unique; under d^{-1} a nonzero image has one predecessor and the collision image zero has exactly the two predecessors $\{0, 6\}$. Hence each reverse level lists the full predecessor set. The fixed-endpoint and remaining-accounting bounds give the finite depth cap. Moreover, an $\mathbf{E}/\mathbf{R}/\mathbf{N}/\mathbf{A}$ exclusion is permanent under further reverse extension: an early fusion cannot be undone, a role violation contradicts the prescribed labelled trace, a shorter word to the fixed endpoint remains shorter, and added prefix length cannot repair an exceeded debt/length bound. Induction on reverse depth therefore proves both termination and exhaustion; no unlisted admissible leaf exists.

Back-Solving Exhaustion Lemma. The inverse recurrence above has exactly the source-addressed surviving leaves displayed below. Every other inverse leaf ends at its first $\mathbf{E}$, $\mathbf{R}$, $\mathbf{N}$, or $\mathbf{A}$ exclusion. A repeated word in two rows denotes two different labelled corridor sources, not a quotient.

cell	c/role	cap	surviving prefixes
132/2, (3, I)	2/D1	3	011 -> X1
132/2, (3, I)	1/D2	5	00001 -> X2
132/2, (3, I)	2/D2	6	00001 -> X3; 00001 -> X4; 000101 -> X5; 101001 -> X5; 010101 -> X6
213/2, (3, I)	1/D1	2	01 -> X8
213/2, (3, I)	2/D2	2	1 -> X7; 01 -> X9
213/2, (3, S)	2/D2	2	01 -> X10
213/2, (4, I)	1/D1	4	01 -> X13; 0101 -> X17
213/2, (4, I)	2/D1	3	1 -> X11; 1 -> X12; 01 -> X15; 101 -> X16
213/2, (4, I)	2/D2	2	01 -> X14

All omitted $c/role$ combinations have empty survivor sets under the same recurrence. The cellwise checksum is

$$(7 \rightarrow 6) + (3 \rightarrow 3) + (1 \rightarrow 1) + (7 \rightarrow 7) = 18 \rightarrow 17, \quad (\text{BT-count})$$

where only X5 identifies two leaves. That identification is legal because the two prefixes induce the same typed post-tag placement and the same full normalization/accounting gate semantics. In contrast, no mass-endpoint quotient is used here.

Expanding the 17 typed leaves gives the following prefix-solution table. The prefix set is part of the equality: two prefixes may be identified only when they induce the same displayed typed state and the same normalization/accounting semantics.

The state identities and prefix data are:

id	negative cell	$c/role$	marked-prefix set	k
X1	132/2, (3, I)	2/D1	{011}	3
X2	132/2, (3, I)	1/D2	{00001}	5
X3	132/2, (3, I)	2/D2	{00001}	5
X4	132/2, (3, I)	2/D2	{00001}	5
X5	132/2, (3, I)	2/D2	{000101, 101001}	6
X6	132/2, (3, I)	2/D2	{010101}	6
X7	213/2, (3, I)	2/D2	{1}	1
X8	213/2, (3, I)	1/D1	{01}	2
X9	213/2, (3, I)	2/D2	{01}	2
X10	213/2, (3, S)	2/D2	{01}	2
X11	213/2, (4, I)	2/D1	{1}	1
X12	213/2, (4, I)	2/D1	{1}	1
X13	213/2, (4, I)	1/D1	{01}	2
X14	213/2, (4, I)	2/D2	{01}	2
X15	213/2, (4, I)	2/D1	{01}	2
X16	213/2, (4, I)	2/D1	{101}	3
X17	213/2, (4, I)	1/D1	{0101}	4

The corresponding typed payloads are:

id	post-tag placement	accounting	corridor source
X1	0:s, 1:D2+F4, 3:D1	(-3, 3, -6)	0:D2+F4, 2:D1, 5:s
X2	0:D1, 1:s, 3:D2, 4:F4	(0, 5, -5)	0:F4, 3:D1, 4:s, 5:D2
X3	0:s, 3:D2, 4:D1+F4	(-2, 5, -7)	0:D1+F4, 2:s, 5:D2
X4	1:s, 3:D2, 4:D1+F4	(-3, 5, -8)	0:D1+F4, 4:s, 5:D2
X5	0:s, 2:D1+F4, 3:D2	(-2, 6, -8)	0:D1+F4, 2:s, 5:D2
X6	2:s, 3:D2, 4:D1+F4	(-2, 6, -8)	0:D1+F4, 2:s, 5:D2
X7	0:D1+F4, 1:D2, 4:s	(-1, 1, -2)	0:D1+F4, 2:D2, 4:s
X8	0:s, 1:D1, 2:F4, 4:D2	(0, 2, -2)	0:F4, 1:D1, 3:D2, 5:s
X9	0:s, 1:D2, 2:D1+F4	(-1, 2, -3)	0:D1+F4, 1:D2, 5:s
X10	0:s, 1:D2, 2:D1+F4	(-1, 2, -3)	0:D1+F4, 1:D2, 5:s
X11	0:D2+F4, 1:D1, 4:s	(-1, 1, -2)	0:D2+F4, 2:D1, 3:s
X12	0:D2+F4, 1:D1, 5:s	(-2, 1, -3)	0:D2+F4, 2:D1, 5:s
X13	0:s, 1:D1, 2:F4, 5:D2	(0, 2, -2)	0:F4, 1:D1, 4:D2, 5:s
X14	1:D2, 2:D1+F4, 3:s	(-1, 2, -3)	0:D1+F4, 1:D2, 3:s
X15	0:s, 1:D1, 2:D2+F4	(-3, 2, -5)	0:D2+F4, 1:D1, 5:s
X16	0:s, 1:D1, 2:D2+F4	(-2, 3, -5)	0:D2+F4, 2:D1, 5:s
X17	0:D2, 1:D1, 2:s, 4:F4	(0, 4, -4)	0:F4, 1:D1, 4:D2, 5:s

Let

$$\mathfrak{B}_{\neq \emptyset} = \{132/2, (3, I); 213/2, (3, I); 213/2, (3, S); 213/2, (4, I)\}.$$

The theorem-facing equality is the disjoint-union statement

$$\bigsqcup_{b \in \mathfrak{B}_{\neq \emptyset}} \text{Sol}_{\text{BT}}(b) = \{\widehat{X}_1, \dots, \widehat{X}_{17}\}$$

over the four nonempty negative cells, together with (**Empty-X**) for the fifth. There are 18 marked-prefix solutions because **X5** has two tied provenances, but exactly 17 typed states. No landing value is used in this elimination.

B.3 Return Admission

For each $q \in Q^{\text{kin}}(\widehat{X})$, let $A_q(\widehat{X})$ be the uniquely determined shallow candidate obtained after terminal-rotation elimination. Define

$$\mathbf{N}(\widehat{X}, q) = 1 \iff A_q(\widehat{X}) \text{ has fixed-endpoint minimum length,}$$

and

$$\mathbf{A}(\widehat{X}, q) = 1 \iff A_q(\widehat{X}) \text{ satisfies the applicable corridor balance and debt bound.}$$

Rank preservation and the prescribed packet roles are already part of Q^{kin} . The admitted relation is therefore

$$Q(\widehat{X}) = \{q \in Q^{\text{kin}}(\widehat{X}) : \mathbf{N}(\widehat{X}, q) = \mathbf{A}(\widehat{X}, q) = 1\}. \quad (\text{Admission})$$

If the marked prefix has length k , a one-return suffix has the forced form

$$p^q d p^{a_q} d, \quad L_q = k + q + a_q + 2, \quad (\text{RA-word})$$

where adjacency uniquely determines a_q . A direct suffix has the form $p^{a_D} d$ and length $L_D = k + a_D + 1$. Write L_* for the shortest length to the same exact mass endpoint. Then

$$\mathbf{N} = 1 \iff L = L_*.$$

For a second-corridor candidate, its surplus is $S_2 = 13 - L$; the typed state supplies $S_{\text{in}} < 0$, and

$$\mathbf{A} = 1 \iff S_{\text{in}} + S_2 \geq 0. \quad (\text{RA-2})$$

For a first-corridor candidate, $S_1 = 7 - L < 0$. Its accounting entry below is **S1**; r/m , where m is the number of role-compatible normalized second corridors and r is the number that repay $-S_1$; here

$$\mathbf{A} = 1 \iff r > 0. \quad (\text{RA-1})$$

Mass endpoints are displayed as seven-coordinate strings. The following is the complete one-return elimination. In the accounting column, an expression $x + y = z$ means $S_{\text{in}} + S_2 = S_{\text{total}}$. The decision column encodes the two gates: **in Q** or **direct** means $(N, A) = (1, 1)$, **N** means $(0, 1)$, **A** means $(1, 0)$, and **N+A** means $(0, 0)$.

X	c	q	a_q	endpoint	L/L_*	accounting	decision
X1	2	0	5	6000010	10/8	-3+3=0	N
X1	2	1	3	6000100	9/9	-3+4=1	in Q
X1	2	5	6	6000100	16/9	-3-3=-6	N+A
X2	1	1	2	4020010	10/10	S1=-3; 4/12	in Q

X	c	q	a_q	endpoint	L/L_*	accounting	decision
X2	1	4	6	4020100	17/9	S1=-10; 0/8	N+A
X3	2	1	2	6010000	10/9	-2+3=1	N
X3	2	4	6	6010000	17/9	-2-4=-6	N+A
X3	2	6	4	6000100	17/11	-2-4=-6	N+A
X4	2	1	2	6000010	10/10	-3+3=0	in Q
X4	2	4	6	6000100	17/9	-3-4=-7	N+A
X4	2	6	4	6000100	17/9	-3-4=-7	N+A
X5	2	0	4	6000100	12/11	-2+1=-1	N+A
X5	2	2	2	6000010	12/10	-2+1=-1	N+A
X5	2	5	6	6000100	19/11	-2-6=-8	N+A
X6	2	1	2	6000100	11/11	-2+2=0	in Q
X6	2	6	4	6000010	18/10	-2-5=-7	N+A
X7	2	1	5	6001000	9/8	-1+4=3	N
X7	2	3	3	6001000	9/8	-1+4=3	N
X7	2	4	2	6000100	9/9	-1+4=3	in Q
X8	1	0	5	4200010	9/8	S1=-2; 4/9	N
X8	1	2	3	4002100	9/9	S1=-2; 0/7	A
X8	1	3	2	4200010	9/8	S1=-2; 4/9	N
X9	2	0	5	6000010	9/8	-1+4=3	N
X9	2	2	3	6000100	9/7	-1+4=3	N
X9	2	3	2	6000010	9/8	-1+4=3	N
X10	2	0	5	6000100	9/8	-1+4=3	N
X10	2	3	2	6000100	9/8	-1+4=3	N
X11	2	1	5	6000100	9/8	-1+4=3	N
X11	2	3	3	6000100	9/8	-1+4=3	N
X12	2	1	5	6000010	9/9	-2+4=2	in Q
X12	2	3	3	6000010	9/9	-2+4=2	in Q
X13	1	0	5	4000210	9/8	S1=-2; 2/6	N
X13	1	2	3	4001200	9/7	S1=-2; 7/10	N
X14	2	0	5	6100000	9/7	-1+4=3	N
X14	2	2	3	6010000	9/8	-1+4=3	N
X15	2	0	5	6000010	9/8	-3+4=1	N
X15	2	2	3	6001000	9/7	-3+4=1	N
X16	2	0	5	6000010	10/9	-2+3=1	N
X16	2	2	3	6001000	10/7	-2+3=1	N
X17	1	0	4	4002010	10/10	S1=-3; 2/6	in Q

The direct candidates are exhausted by the same formulas with no return exponent:

x	c	a_D	endpoint	L/L_*	accounting	decision
x2	1	3	4020100	9/9	S1=-2; 3/8	direct
x3	2	3	6010000	9/9	-2+4=2	direct
x4	2	3	6000100	9/9	-3+4=1	direct
x5	2	4	6000100	11/11	-2+2=0	direct
x6	2	3	6000010	10/10	-2+3=1	direct
x7	2	6	6001000	8/8	-1+5=4	direct
x8	1	5	4200010	8/8	S1=-1; 4/9	direct
x9	2	5	6000010	8/8	-1+5=4	direct
x10	2	5	6000100	8/8	-1+5=4	direct
x11	2	6	6000100	8/8	-1+5=4	direct
x12	2	6	6001000	8/7	-2+5=3	N
x13	1	5	4000210	8/8	S1=-1; 4/6	direct
x14	2	5	6010000	8/8	-1+5=4	direct
x15	2	5	6000010	8/8	-3+5=2	direct
x16	2	5	6000010	9/9	-2+4=2	direct

Thus the rowwise equality (Admission) gives

id	Q^kin(X)	admitted Q(X)	direct
x1	{0,1,5}	{1}	none
x2	{1,4}	{1}	admitted
x3	{1,4,6}	empty	admitted
x4	{1,4,6}	{1}	admitted
x5	{0,2,5}	empty	admitted
x6	{1,6}	{1}	admitted
x7	{1,3,4}	{4}	admitted
x8	{0,2,3}	empty	admitted
x9	{0,2,3}	empty	admitted
x10	{0,3}	empty	admitted
x11	{1,3}	empty	admitted
x12	{1,3}	{1,3}	rejected by N
x13	{0,2}	empty	admitted
x14	{0,2}	empty	admitted
x15	{0,2}	empty	admitted
x16	{0,2}	empty	admitted
x17	{0}	{0}	none

Summing the rows gives

$$40 = 8_{\text{admitted}} + 21_N + 1_A + 10_{N+A}, \quad (\text{Q-count})$$

and the direct candidates give

$$15 = 14_{\text{admitted}} + 1_N. \quad (\text{D-count})$$

No return candidate passes both N and A while remaining outside $Q(\widehat{X})$. No direct candidate passes its declared gate while remaining outside the direct relation. Thus endpoint normalization and accounting are the complete admission interface **on this five-cell surface**. This statement is not promoted to other ranks or ambient sizes.

B.4 Heavy-Pair Exit Exhaustion

For a typed rank-three state Y , define

$$\mathcal{E}(Y) = \{w : w \text{ is an admissible endpoint-normalized strict heavy-pair exit from } Y\},$$

and

$$\Gamma(Y, t) = \{w(t) : w \in \mathcal{E}(Y)\}. \quad (\text{Gamma-image})$$

Words use $0 = p, 1 = d$ and act left to right. Completeness is obtained before the singleton is evaluated. On the finite rank-three mass graph, let $\delta_Y(M)$ be the shortest rank-preserving distance from the mass placement of Y to M . Retain a parent edge $M \xrightarrow{a} M'$ exactly when

$$\delta_Y(M) + 1 = \delta_Y(M'), \quad (\text{Exit-DAG})$$

and retain a strict terminal edge to Z exactly at the minimum distance to that exact endpoint Z . Backtracking this shortest-parent DAG gives every tied endpoint-shortest exit word. A terminal parent-role check then selects the exits fusing the two heavy packet identities. Finally, the displayed debt bound discards the overlength exits. No singleton landing is read in any of these three operations.

The resulting equalities are below. An accepted entry records (word, length, target, t, S_2); its last two fields are obtained only by passive evaluation. The over-debt relation lists every other endpoint-shortest exit that fuses the correct heavy pair, together with its length and target.

The admitted relation and its passive images are:

id	defect / typed Y	debt / max ell_2	complete $E(Y)$	Gamma(Y, t)
Y1	0132450; 0:D2+F4,2:D1,4:s	2 / 11	0010001 [7,6010000,2,6]; 010100001 [9,6000100,4,4]; 0101001001 [10,6000010,5,3]	{2,4,5}
Y2	0132450; 0:D2+F4,2:D1,5:s	3 / 10	0010001 [7,6010000,2,6]; 010001001 [9,6000100,4,4]; 011010001 [9,6000100,4,4]; 01000001 [8,6000010,5,5]	{2,4,5}
Y3	0213450; 0:D1+F4,1:D2,5:s	1 / 12	00001001 [8,6001000,3,5]; 0000001 [7,6000100,4,6]; 00010001 [8,6000010,5,5]; 01000001 [8,6000010,5,5]	{3,4,5}
Y4	0214350; 0:D1+F4,4:D2,5:s	1 / 12	011000100001 [12,6010000,2,1]; 011010000001 [12,6010000,2,1]; 10100001 [8,6001000,3,5]; 101010001 [9,6000010,5,4]	{2,3,5}
Y5	0214350; 0:D1+F4,3:D2,5:s	3 / 10	0100001 [7,6001000,3,6]; 01010001 [8,6000010,5,5]	{3,5}

For the checksum below, write

$$(a, o, w) = (\text{admitted}, \text{over-debt}, \text{wrong-pair}).$$

The rejected correct-pair exits and complete endpoint-shortest checksums are:

id	split (a, o, w)
Y_1	(3, 5, 16)
Y_2	(4, 8, 13)
Y_3	(4, 5, 15)
Y_4	(4, 2, 17)
Y_5	(2, 4, 16)

The complete over-debt relation is:

id	word	length	endpoint
Y_1	00000010000001	14	6100000
Y_1	000000100001001	15	6001000
Y_1	000000110000001	15	6001000
Y_1	00010101000001	14	6100000
Y_1	000101010001001	15	6001000
Y_2	0001010010001	13	6100000
Y_2	00010100101001	14	6001000
Y_2	00010101000001	14	6001000
Y_2	1000101000001	13	6100000
Y_2	10001010001001	14	6001000
Y_2	1010010010001	13	6100000
Y_2	10100100101001	14	6001000
Y_2	10100101000001	14	6001000
Y_3	000001010100001	15	6100000
Y_3	0000101000100001	16	6010000
Y_3	100000010100001	15	6100000
Y_3	100101010000001	15	6100000
Y_3	1001010110100001	16	6010000
Y_4	0110101000001	13	6100000
Y_4	0110110101001	13	6000100
Y_5	001000100001	12	6010000
Y_5	001010000001	12	6010000
Y_5	0010101000001	13	6100000
Y_5	0010110101001	13	6000100

Thus each row proves a word-set equality, not merely the existence of enough witnesses. Across the five rows the complete rank-three relation contains 17 admitted heavy-pair exits, 24 correct-pair exits rejected only by the debt bound, and 77 tied endpoint-shortest exits rejected by the terminal packet roles. The singleton images in Γ are derived images of the admitted word sets.

B.5 Derived Negative Spectra

Composing the first-corridor images with Γ and adjoining the local second-corridor images gives:

cell	direct spectrum	one-return spectrum
132/2, (3, I)	{2, 4, 5}	{2, 4, 5}
213/2, (3, I)	{3, 4, 5}	{4}
213/2, (3, S)	{4}	empty
213/2, (4, I)	{2, 3, 4, 5}	{3, 5}
321/3, (3, I)	empty by (Empty-X)	empty by (Empty-X)

No displayed spectrum contains offset one. This final table is a set-image calculation; all universal work is confined to Back-Solving, Return Admission, and Heavy-Pair Exit Exhaustion above.

Appendix C Computational Artifacts

All listed artifacts are available in the [RIME repository](#) under `experiments/paper27/`; short paths are relative to that directory.

artifact surface	role	short path
theorem-facing records	fixed-scope classifications and symbolic tables	<code>results/</code>
producers	entry-section, relation, return, and exit enumeration	<code>*.py</code>
validators	formula checks, source binding, and closure checks	<code>validation/</code>
release manifest	ordered exact-byte closure and closure-class policy	<code>release-manifest.json</code>
validation receipt	local closure verification, excluded from its own closure	<code>results/*validation-receipt.json</code>

The release identity contains only the canonical manuscript and theorem-facing closure. Package-only, historical, and receipt artifacts do not alter theorem identity.

The receipt records local closure verification, not independent mathematical validation. Replay flags state which producers were re-executed for that receipt; a bound artifact digest does not by itself establish theorem truth.

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